

Decision Support System for Cancer Chemotherapy

Decide the right dose, before side effects appear.

WHAT THE CLINICIAN SEES



THE PLAN THE CLINICIAN REVIEWS · 9 DOSES OVER 120 DAYS



The whole course of treatment is laid out before the first dose — how the tumour is expected to respond, and how close each dose comes to every organ's safe limit. Here 4 of 9 doses were lowered in advance, because the lung would otherwise have been pushed past its limit.

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PROBLEM

Chemotherapy has to shrink the tumour without harming healthy organs — but the dose is usually lowered only **after** the patient already has side effects.

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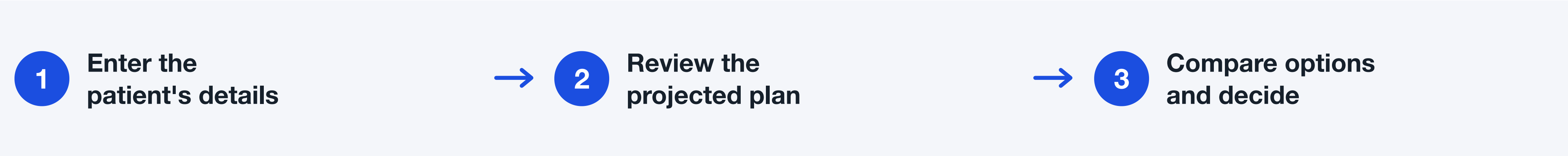
SOLUTION

Before a dose is given, the clinician can see what it is expected to do to the tumour and to **each major organ** — and which organ is limiting it.

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IMPACT

The full course can be reviewed and alternatives compared **before treatment starts**, so the dose is decided in advance rather than in reaction.



13

organs checked before every single dose is given

4 of 9

doses lowered in advance, before any side effect had appeared

28

alternative schedules compared before treatment begins

TEAM & INSTITUTION

Author names — Institution / Department

Simulation output, for research and evaluation. Not a medical device and not for clinical decision-making. Organ safety limits shown are illustrative.

